

CONTACT

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- github.com/Togi1

TECHNICAL SKILLS

Programming & Architecture
C# (OOP, ScriptableObjects, Physics & Input Systems), Python (Data Structures, Algorithms)

Game Engine & Systems
Unity 2D/3D, Mechanic Prototyping, Game Feel (Juice), Progression & In-Game Economy Loops

Production & Pipeline
Git / GitHub, Unity Cloud, Jira (Agile/Sprint Tracking), Steamworks Distribution

3D & Hardware Engineering
Blender (Low/Mid-Poly, UV, Rigging), Level Design, Sensor Robotics & Hardware Integration

LANGUAGES

English: Professional Working (B2)
Turkish: Native

FOCUS & INTERESTS

Gameplay Mechanics, System Design, Game Economics, Rapid Prototyping, Licensed Badminton Athlete.

TOLGAHAN AY

Gameplay Developer & Technical Game Designer

PROFESSIONAL PROFILE / SUMMARY

Forward-thinking Game Developer and Digital Game Design undergraduate with a disciplined engineering background marked by ranking **first in high school** and holding **two registered patents**. Specializes in building modular, data-driven gameplay architectures in **Unity (C#)** using ScriptableObjects and OOP principles. Adept at translating creative concepts into playable, polished prototypes—combining responsive physics-based player controls, engaging game feel (juice), and balanced progression economies within collaborative agile pipelines (Git, Unity Cloud, Jira).

FEATURED GAME PROJECTS / ENGINEERING

Project PO535-T — Game Designer & PM — **In Active Dev (PC)**

System Design | Diagnostic Matrix | Project Management | Milestone Gates

- Authoring the comprehensive Game Design Document (GDD) and managing a 4-person multidisciplinary team across a 5-phase production roadmap with strict Go/No-Go milestone gates.
- Engineered core diagnostic rule systems, symptom-masking deception algorithms, and tactical workbench mechanics inspired by *Papers, Please*.

2D Pixel Art Fishing Game — Gameplay & Systems Designer — **Live Demo: togi1.github.io**

Unity | C# | ScriptableObjects | Economy Loop

- Architected a decoupled, data-driven inventory and loot drop system using **ScriptableObjects**, enabling non-technical scaling of fish rarities and attributes.
- Engineered core gameplay progression and shop economy, optimizing player retention loops through procedural fishing tension mechanics and equipment upgrade paths.

3D Mobile Dart Game — Gameplay Developer — **Live Demo: togi1.github.io**

C# | Physics Engine | Touch Input | Game Feel

- Programmed high-precision projectile physics calibrated for multi-touch mobile input, supported by a dynamic vector-based zone scoring engine.
- Integrated tactile game feel (juice): directional camera shake, haptics, and particle bursts upon impact.

BlackFriday — Pipeline & Gameplay Contributor — **Multi-Disciplinary Team**

Unity Cloud | Jira | Steamworks Pipeline

- Collaborated in agile sprint cycles, managing multi-branch version control via **Unity Cloud**, Jira task tracking, and Steamworks deployment packaging.

REGISTERED PATENTS & APPLIED R&D / TURKPATENT

Autonomous Line-Tracking Robotics Platform [TR2022017364U5]
Engineered sensor-driven micro-controller navigation algorithms and closed-loop motor control hardware for precision automated tracking.

Multi-Functional Storage & Peripheral Hardware Unit [TR2022018474A2]
Designed ergonomic 3D spatial casing integrated with USB/SD modular interfaces and custom mechanical locking mechanisms.

EDUCATION

Muğla Sıtkı Koçman University
BSc in Digital Game Design
Expected 2025 – Present

High School Graduation
Graduated with 1st Degree Honors (Valedictorian)

CERTIFICATIONS

- **Foundational C# Certification** — Microsoft Authorized
- **Robotic Coding Specialist** — Advanced Embedded Systems
- **DiscoverEU Laureate** — European Commission (Mobility Grant)